

Supplementary S1: Rule Redundancy and Stability Analysis of the Auditor's Rationale Layer

Related main-paper sections: §3.3, §5.5, §5.9, §6.1, §6.2(b)

S1.1 Introduction

Note: the stability analysis operates on the ablation arm (§4.5), where the upstream judge verdict has been replaced with random noise and its reasoning trace replaced with a content-free monologue. This makes the test deliberately harsh — it measures which rules survive when the judge signal is maximally degraded.

Section 5.5 of the main paper reports that the Correspondence Auditor's binary verdict is highly reproducible ($ICC = 0.979$) but its rule-citation rationale is not (Krippendorff's $\alpha = 0.238$). The present supplementary reports the exploratory post-hoc analysis that investigates the structure of that instability.

The central question is whether the low α reflects **label-substitution instability** (the auditor detects the same underlying concept but cites different, semantically-equivalent rules across runs) or **presence/absence instability** (the entire concept family flickers on and off). The distinction matters because label substitution is addressable through rule compression — merging synonymous rules into a single cluster — whereas presence/absence flicker is a deeper instability that compression cannot resolve.

A note on comparability with the main paper: the per-rule flicker rates reported in §5.5 are computed on the $k = 6$ frame at the individual-rule level; the cluster-level rates reported below are computed on the $k = 5$ rerun-only frame (§S1.2.1). The two sets of figures are consistent but are not directly comparable cell-by-cell.

S1.2 Method

S1.2.1 Data

The analysis uses the same ablation-arm test-retest data described in §4.7 of the main paper: 297 cases with $k = 6$ observations each (one seed pass plus five reruns), yielding 1,781 unique observations over the 14 observed rules. The incidence matrix is case $\times$ pass $\times$ rule $\rightarrow \{0, 1 \text{ fired}\}$, constructed from raw audit JSONs and verified against the author's pipeline through provenance-chain tests (repository `tests/`).

Two analysis frames are used, following the conventions of the source analysis (Tsereteli, 2026). Alpha coefficients are computed on the $k = 6$ frame (297 cases: seed + 5 reruns) to match the main paper's §5.5 convention. Clustering and cluster-level flicker are computed on the $k = 5$ rerun-only balanced panel (296 complete cases), the frame used throughout the source analysis. The frame applicable to each table is stated in its caption.

S1.2.2 Analysis Overview

The analysis proceeds in three stages:

1. **Pairwise co-firing and clustering.** Co-firing patterns, L1-regularised logistic models, case-level fire-rate correlations, conditional redundancy within verdict/ground-truth/quadrant strata, and hierarchical clustering were used to identify natural rule groups. Full methodological details are in Tsereteli (2026).
2. **Compressed rulebook evaluation.** Five compressed rulebooks were constructed by merging identified clusters. MASI Krippendorff's α and case-level reproducibility metrics were recomputed for each.
3. **Cluster-level flicker analysis.** To distinguish label-substitution from presence/absence instability, we measured: when any member of a compressed cluster fires in a case, how often does the cluster fire in *all five* reruns?

All statistics are reported rounded to three decimal places; unrounded values are available in the source CSVs

(`rulebook-redundancy-analysis/results/compressed_rulebook/`).

S1.2.3 Rule Clusters Identified

The analysis converges on a consistent natural geometry across all methods:

Cluster	Rules	Strongest Evidence
Tight J core	J3, J4, J5	Jaccard 0.78–0.87; case-level Pearson 0.91–0.95
Nearby J pair	J1, J2	Jaccard 0.75; case-level Pearson 0.91
Broad J block	J1–J5	Fire-rate clusters at dissimilarity threshold 0.20
Factual cluster	F1, F2	Jaccard 0.80; case-level Pearson 0.88
Universal cluster	U1, U3	Jaccard 0.82; weaker case-level correlation

These clusters are consistent across pass-level Jaccard, case-level fire-rate correlation, sparse logistic models, and conditional redundancy within strata.

S1.2.4 Compressed Rulebooks

Spec	Labels	Merges Applied
original	14	No merges
j_only	11	J1/J2 → J12; J3/J4/J5 → J345
natural_tight	9	F1/F2 → F12; J1/J2 → J12; J3/J4/J5 → J345; U1/U3 → U13; U2, F3, H1, S1, U4 retained
j_all	10	J1/J2/J3/J4/J5 → J_ALL
broad_clusters	6	F1/F2/F3/U1/U3 → F123U13; J1–J5 → JALL; H1, S1, U2, U4 retained

Three further J-family-only specs (*jfamilyoriginal*, *jfamilytight*, *jfamilyall*) restrict the rulebook to the J family at increasing levels of compression; these isolate the J-family signal and appear in Table S1.1.

S1.3 Results

S1.3.1 Compressed Alpha

The best compression raises α from 0.238 to 0.278 — a real improvement (paired bootstrap 95% CI for the increase: [+0.031, +0.047]; 3,000 case-level replicates, seed 12345; Tsereteli, 2026), but still far below a stable rationale layer.

The J-family-only rows isolate the judge-process signal. Within this series α rises with compression (0.100 → 0.112 → 0.128), so compression does remove some label

fragmentation — but it starts near zero and ends near zero. The endpoint is instructive: under *jfamilyall*, where the entire J family is collapsed to a single binary label, $\alpha = 0.128$ is an *optimistic* figure, because 63.6% of cases carry no J citation on any run and therefore contribute empty-set agreement (cf. the upper-bound caveat in §5.5 of the main paper). Even under maximal coarsening and maximal empty-agreement inflation, the J-family signal does not reproduce. (α values across specs with different label universes are not commensurate in magnitude — expected disagreement depends on the label space — so we do not rank *jfamilyall* against the full-rulebook specs; the comparison that matters is within the J-family series.)

Table S1.1 — MASI Krippendorff's α by Compressed Rulebook Spec

Spec	Labels	Cases	Runs	α (MASI)	Obs. Disagreement	Exp. Disagreement
original	14	297	6	0.238	0.583	0.765
natural_tight	9	297	6	0.278	0.534	0.739
j_only	11	297	6	0.242	0.576	0.760
j_all	10	297	6	0.244	0.573	0.758
broad_clusters	6	297	6	0.263	0.495	0.672
jfamilyoriginal	5	297	6	0.100	0.156	0.173
jfamilytight	2	297	6	0.112	0.152	0.171
jfamilyall	1	297	6	0.128	0.147	0.168

Computed on the $k=6$ frame (297 cases: seed + 5 reruns). Table generated by `generate_redundancy_tables.py` from verified CSV data; row order and values are byte-identical to the generated artifact.

S1.3.2 Cluster-Level Flicker

This is the decisive diagnostic. If the J-family instability is label substitution, merging J3/J4/J5 into J345 should produce a stable cluster. If it is presence/absence flicker, the merged cluster should itself flicker.

Table S1.2 — Cluster-Level On/Off Stability (natural_tight spec)

Cluster	Members	Cases Seen	Fires All 5	Flicker Rate	Mean Fire Rate When Seen
F12	F1 F2	198	192	0.030	0.988
F3	F3	144	4	0.972	0.442
H1	H1	259	18	0.931	0.486
J12	J1 J2	92	1	0.989	0.287
J345	J3 J4 J5	59	1	0.983	0.281
S1	S1	115	96	0.165	0.894
U13	U1 U3	294	217	0.262	0.899
U2	U2	2	0	1.000	0.200
U4	U4	282	20	0.929	0.540

Computed on the $k=5$ frame (296 complete cases: reruns only). Table generated by `generate_redundancy_tables.py` from verified CSV data; row order and values are byte-identical to the generated artifact. U2 is seen in only 2 cases; its flicker rate is not interpretable and is reported for completeness only.

F12 behaves like a genuinely stable merged concept: once the F1/F2 family appears, it appears in all five reruns 97.0% of the time. **J12** and **J345** do not: they flicker on 98.9% and 98.3% of the cases they appear in, respectively. The J-family instability is not label substitution — it is presence/absence flicker.

The non-J high-flicker clusters confirm that the instability is not confined to judge-process rules. F3 (flicker rate 0.972), H1 (0.931), and U4 (0.929) are similarly unstable.

S1.3.3 Summary

The natural clustering is genuine: J3/J4/J5 co-fire at Jaccard 0.78–0.87, their case-level fire-rate profiles correlate at Pearson 0.91–0.95, and merging them into J345 removes real label fragmentation. But the merged signal itself flickers. The auditor detects the J345 concept in 59 of 296 cases, yet in only 1 of those 59 cases does it cite the concept across all five reruns. The instability is located in the presence/absence decision itself — whether the concept fires at all — not in the choice among labels.

S1.4 Limitations

1. **Exploratory and post-hoc.** This analysis was not pre-registered. All findings are hypothesis-generating, not confirmatory.
2. **Ablation-arm reruns only.** The reruns process cases without a genuine judge trace (§4.7). This is an important context for interpreting the J-rule results: rules that evaluate the judge's reasoning process (J1–J5) are expected to flicker on the ablation arm because there is no genuine judge reasoning to assess. The architecturally significant finding is that calibrated, fact-grounded rules ($F1/F2 \rightarrow F12$) remain stable under this worst case, demonstrating that Gate B grounding — rather than rule content alone — is the stabilising mechanism.
3. **Five reruns is narrow.** The flicker rate is estimated from $k = 5$ observations per case. The bootstrap noise margin for α is ± 0.02 ; cluster-level flicker-rate estimates carry comparable uncertainty, and small-cluster estimates (e.g., U2, seen in 2 cases) are not interpretable.
4. **J-family flicker is descriptive, not causal.** We do not establish why the J-family signal flickers — whether it reflects genuine non-determinism in the auditor's reasoning, sensitivity to minor prompt-level surface variation, or a property of the underlying LLM's sampling. The explanation matters for remediation but does not change the descriptive finding.
5. **Single domain.** All findings are on the Apollo corpus under one Truth Cartridge. Whether the same cluster geometry and flicker profile hold on other domains is untested.

S1.5 Implications for the Main Paper

Coarsening is insufficient. The most natural response to $\alpha = 0.238$ is to compress the rulebook. We tested this systematically and found it achieves only $+0.04 \alpha$ units. The problem is not label fragmentation; it is that the underlying concept detectors flicker on and off.

The §3.3 interpretability claim is correctly bounded. The main paper states that "individual policy-rule citations should be read at the aggregate-frequency level rather than as stable per-case attributions." This analysis confirms and sharpens that bound: even compressed rule clusters (J12, J345) are not stable per-case attributions. The stable layer is the verdict ($ICC = 0.979$) and the factual-rule cluster F12 (97.0% all-five agreement), not the judge-process (J) or hygiene (H/U) clusters.

Future work direction is clarified. The exploratory findings reframe the §6.2(b) future work trajectory for the rationale layer. The original direction — whether per-rule firing frequency, averaged over stochastic passes, carries a stable signal — is superseded by the present

analysis: the instability is presence/absence flicker, not label substitution, so aggregating unstable presence/absence decisions across passes cannot rescue a fundamentally ungrounded predicate. The F12 stability result points at the correct direction: extending Gate B's decomposed, source-grounded evaluation to J-family predicates, so that every rule receives the grounding that currently stabilises F1/F2. This is the architectural motivation for the v3 redesign.

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Reference:

Tsereteli, G. (2026). Rule Redundancy and Stability Analysis: Exploratory Report. Technical report accompanying LLM-as-Judge Deception Audit. Study repository:

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